

PRAMISON (Pramipexole Tablets 0.125mg)**PRAMISON (Pramipexole Tablets 0.25mg)****PRAMISON (Pramipexole Tablets 0.1mg)**

Each tablet in the form of uncoated tablet.

Description

Pramison Tablets 0.125mg (Pramipexole 0.088mg)

White to off-white, round, flat, bevelled edge uncoated tablets, debossed with 'Y' on one side '41' on the other side with the presence of score line and the dimension of the tablet is 5.5mm.

Each uncoated tablet contains Pramipexole Dihydrochloride Monohydrate 0.125 mg.

Pramison Tablet 0.25mg (Pramipexole 0.18mg)

White to off-white, oval, biconcave, bevelled edge uncoated tablets, debossed with 'Y' and '42' separated by score line on one side and plain with score line on other side and the dimension of the tablet is 6.5mm x 4.5mm

Each uncoated tablet contains Pramipexole Dihydrochloride Monohydrate 0.25 mg

Pramison Tablet 1mg (Pramipexole 0.7mg)

White to off-white, round, flat, bevelled edge uncoated tablets debossed with 'Y' and '45' separated by score line on one side and plain with score line on other side and the dimension of the tablet is 9.3mm.

Each uncoated tablet contains Pramipexole Dihydrochloride Monohydrate 1 mg

Pharmacodynamics

Pramipexole, active ingredient of PRAMISON is a dopamine-agonist and binds with high selectivity and specificity to the dopamine D2-subfamily receptors and has a preferential affinity to D3-receptors; it has full intrinsic activity. PRAMISON alleviates parkinsonian motor deficits by stimulation of dopamine receptors in the striatum. Pramipexole protect dopamine neurons from degeneration in response to ischaemia or methamphetamine neurotoxicity. The precise mechanism of action of PRAMISON as a treatment for restless leg syndrome (RLS) is not known. Although the pathophysiology of RLS is largely unknown, neuropharmacological evidence suggests primary dopaminergic system involvement. Positron emission tomographic studies suggest that a mild striatal presynaptic dopaminergic dysfunction may be involved in the pathogenesis of RLS.

Pharmacokinetics

Pramipexole is rapidly and completely absorbed following oral administration. The absolute bioavailability is >90% and the maximum plasma concentration occur between 1 and 3 hours. The rate of absorption is reduced by food intake but not the overall extent of absorption. Pramipexole shows linear kinetics and a relatively small interpatient variation of plasma level. In humans, the protein-binding of Pramipexole is very low (<20%) and the volume of distribution is large (400 L). High brain tissue concentrations were observed in the

rat (approximately 8-fold compared to plasma). Pramipexole is metabolised in man only to a small extent.

Renal excretion of unchanged Pramipexole is the major route of elimination and accounts for about 80% of dose. Approximately 90% of a ^{14}C -labelled dose is excreted through the kidneys while <2% is found in the feces. The total clearance of Pramipexole is approximately 500 mL/min and the renal clearance is approximately 400 mL/min. The elimination half-life ($t_{1/2}$) varies from 8 hrs in the young to 12 hrs in the elderly.

Absorption

Rapidly and completely absorbed. Bioavailability: >90%. Time to peak plasma concentration: 1 to 3 hr.

Distribution:

Volume of distribution: 400 L. protein-binding of pramipexole is very low (<20%)

Metabolism:

Only to a small extent.

Excretion:

Via urine (approx. 90%) as unchanged drug).

Elimination half-life:

8 hrs in the young to 12 hrs in the elderly

Age:

Pramipexole clearance decreases with the age as the $t_{1/2}$ and clearance are about 40% longer and 30% lower respectively, in elderly (≥ 65 years) compared with young healthy volunteers (<40 years). The difference is most likely due to well-known reduction in renal function with age, since Pramipexole clearance is correlated with renal function, as measured by creatinine clearance

Indication

Pramipexole is indicated for treatment of the signs and symptoms of advanced idiopathic Parkinson's disease. Pramipexole may be used as monotherapy or combination with levodopa. Symptomatic treatment of idiopathic restless leg syndrome.

Recommended Dosage.

Dosage and Administration

(all dose information refers to pramipexole salt form).

Parkinson's disease

The tablets should be taken orally, swallowed with water, and can be taken either with or without food. The daily dosage is administered in equally divided doses 3 x per day.

Initial treatment:

As shown below dosages should be increased gradually from a starting-dose of 0.375 mg per

day and then increased every 5 - 7 days. Providing patients do not experience intolerable side-effects, the dosage should be titrated to achieve a maximal therapeutic effect

Ascending – Dose Schedule of Pramison		
Week	Dosage (mg)	Total Daily Dose (mg)
1	3 x 0.125	0.375
2	3 x 0.25	0.75
3	3 x 0.5	1.50

if a further dose increase is necessary the daily dose should be increased by 0.75 mg at weekly intervals up to a maximum dose of 4.5 mg per day.

Maintenance treatment:

The individual dose of pramipexole should be in the range of 0.375 mg to a maximum of 4.5 mg per day. During dose escalation in pivotal studies both, in early and advanced disease efficacy was observed starting at a daily dose of 1.5 mg. This does not preclude that in individual patients doses higher than 1.5mg per day can result in additional therapeutic benefit.

This applies particularly to patients with advanced disease where a reduction of the levodopa therapy is intended.

Treatment discontinuation:

Pramison tablets should be tapered off at a rate of 0.75 mg per day until the daily dose has been reduced to 0.75 mg. Thereafter the dose should be reduced by 0.375 mg per day.(See special warning and precautions)

Dosing in patients with concomitant levodopa therapy.

In patients with concomitant levodopa therapy it is recommended that the dosage of levodopa is reduced during both dose escalation and maintenance treatment with Pramison. This may be necessary in order to avoid excessive dopaminergic stimulation.

Dosing in patients with renal impairment:

The elimination of Pramipexole is dependent on renal function. The following dosage schedule is suggested for initiation of therapy. Patients with a creatinine clearance above 50 ml/min require no reduction in daily dose or dosing frequency.

In patients with a creatinine clearance between 20 and 50 ml/min, the initial daily dose of Pramison tablets should be administered in two divided doses, starting at 0.125 mg twice a day (0.25 mg daily). A maximum daily dose of 2.25 mg pramipexole should not be exceeded. In patients with a creatinine clearance less than 20 ml/min, the daily dose of Pramison tablets

should be administered in a single dose, starting at 0.125 mg daily. A maximum daily dose of 1.5 mg pramipexole should not be exceeded.

If renal function declines during maintenance therapy, reduce Pramison daily dose by same percentage as decline in creatinine clearance, ie. if creatinine clearance declines by 30%, then reduce Pramison daily dose by 30%. The daily dose can be administered in two divided doses if creatinine clearance is between 20 and 50 ml/min, and as a single daily dose if creatinine clearance is less than 20 ml/min

Dosing in patients with hepatic impairment

Dose reduction is not considered necessary in patients with hepatic impairment.

Paediatric population

The safety and efficacy use of Pramison in paediatric patients has not been established.

Restless Legs Syndrome

The tablets should be taken orally, swallowed with water, and can be taken either with or without food. The recommended starting dose of Pramison is 0.125 mg taken once daily 2 - 3 hours before bedtime. For patients requiring additional symptomatic relief, the dose may be increased every 4 - 7 days to a maximum of 0.75 mg per day (as shown in the table below):

Ascending- Dose Schedule of Pramipexole	
Titration Step	Once Daily Evening Dose (mg of salt)
1	0.125
2*	0.25
3*	0.50
4*	0.75
* if needed	

Treatment discontinuation

Pramison can be discontinued without tapered dose reduction. In a 26-week placebo controlled clinical trial, rebound of RLS symptom (worsening of symptom severity as compared to baseline) was observed in 10% of patients after abrupt discontinuation of pramipexole. This effect was found to be similar across all doses.

Dosing in patients with renal impairment

The elimination of Pramison is dependent on renal function and closely related to the creatinine clearance. Patients with a creatinine clearance above 20 ml/min require no reduction in daily dose. The use of Pramison in RLS patients with renal impairment has not been studied.

Dosing in patients with hepatic impairment

Dose reduction is not considered necessary in patients with hepatic impairment, as approx. 90% of absorbed drug is excreted through the kidneys.

Dosing in Children and Adolescent

Safety and efficacy of Pramison have not been established in children and adolescents up to 18 years.

Route of Administration

Oral

Contraindications

Pramipexole tablets are contraindicated in patients with hypersensitivity to Pramipexole or to any of the Excipients.

Warnings and Precautions

When prescribing Pramison in a patient with Parkinson's disease with renal impairment a reduced dose is suggested in line with posology and administration.

Hallucinations

Hallucinations are known as a side-effect of treatment with dopamine agonists and levodopa. Patients should be informed that (mostly visual) hallucinations can occur.

Dyskinesias

In advanced Parkinson's disease, in combination treatment with levodopa, dyskinesias can occur during the initial titration of Pramison. If they occur, the dose of levodopa should be decreased.

Sudden onset of sleep and somnolence

Pramipexole has been associated with somnolence and episodes of sudden sleep onset, particularly in patients with Parkinson's disease. Sudden onset of sleep during daily activities, in some cases without awareness or warning signs, has been reported uncommonly. Patients must be informed of this and advised to exercise caution while driving or operating machines during treatment with Pramison. Patients who have experienced somnolence and/or an episode of sudden sleep onset must refrain from driving or operating machines. Furthermore, a reduction of dosage or termination of therapy may be considered. Because of possible additive effects, caution should be advised when patients are taking other sedating medication or alcohol in combination with pramipexole.

Impulse control disorders

Patients should be regularly monitored for the development of impulse control disorders. Patients and carers should be made aware that behavioural symptoms of impulse control disorders including pathological gambling, increased libido, hypersexuality, compulsive spending or buying, binge eating and compulsive eating can occur in patients treated with dopamine agonists including pramipexole. Dose reduction/tapered discontinuation should be considered if such symptoms develop.

Mania and delirium

Patients should be regularly monitored for the development of mania and delirium. Patients and carers should be made aware that mania and delirium can occur in patients treated with

pramipexole. Dose reduction/tapered discontinuation should be considered if such symptoms develop.

Patients with psychotic disorders

Patients with psychotic disorders should only be treated with dopamine agonists if the potential benefits outweigh the risks.

Coadministration of antipsychotic medicinal products with pramipexole should be avoided.

Ophthalmologic monitoring

Ophthalmologic monitoring is recommended at regular intervals or if vision abnormalities occur.

Severe cardiovascular disease

In case of severe cardiovascular disease, care should be taken. It is recommended to monitor blood pressure, especially at the beginning of treatment, due to the general risk of postural hypotension associated with dopaminergic therapy.

Neuroleptic malignant syndrome

Symptoms suggestive of neuroleptic malignant syndrome have been reported with abrupt withdrawal of dopaminergic therapy.

Driving and using machine

Patients being treated with Pramison and presenting with somnolence and/or sudden sleep episodes must be informed to refrain from driving or engaging in activities where impaired alertness may put themselves or others at risk of serious injury or death (e.g. operating machines) until such recurrent episodes and somnolence have resolved (see also section on special warnings and special precautions for use)

Interactions with Other Medicaments

Plasma protein binding

Pramipexole is bound to plasma proteins to a very low (< 20%) extent, and little biotransformation is seen in man. Therefore, interactions with other medicinal products affecting plasma protein binding or elimination by biotransformation are unlikely. As anticholinergics are mainly eliminated by biotransformation, the potential for an interaction is limited, although an interaction with anticholinergics has not been investigated. There is no pharmacokinetic interaction with selegiline and levodopa.

Inhibitors/competitors of active renal elimination pathway

Cimetidine reduced the renal clearance of pramipexole by approximately 34%, presumably by inhibition of the cationic secretory transport system of the renal tubules. Therefore, medicinal products that are inhibitors of this active renal elimination pathway or are eliminated by this pathway, such as cimetidine, amantadine, mexiletine, zidovudine, cisplatin, quinine, and procainamide, may interact with pramipexole resulting in reduced clearance of pramipexole. Reduction of the pramipexole dose should be considered when these medicinal products are administered concomitantly.

Combination with levodopa

When Pramison is given in combination with levodopa, it is recommended that the dosage of levodopa is reduced and the dosage of other anti-parkinsonian medicinal products is kept constant while increasing the dose of Pramison.

Because of possible additive effects, caution should be advised when patients are taking other sedating medicinal products or alcohol in combination with pramipexole.

Antipsychotic medicinal products

Coadministration of antipsychotic medicinal products with pramipexole should be avoided, e.g. if antagonistic effects can be expected.

Statement on usage during pregnancy and lactation

Pregnancy

The effect on pregnancy and lactation has not been investigated in humans. Pramison should not be used during pregnancy unless clearly necessary, i.e. if the potential benefit justifies the potential risk to the foetus.

Breast-feeding

As pramipexole treatment inhibits secretion of prolactin in humans, inhibition of lactation is expected. The excretion of pramipexole into breast milk has not been studied in women. The concentration of active substance-related radioactivity was higher in breast milk than in plasma. In the absence of human data, Pramison should not be used during breast-feeding. However, if its use is unavoidable, breast-feeding should be discontinued.

Fertility

No studies on the effect on human fertility have been conducted. In animal studies, pramipexole affected oestrous cycles and reduced female fertility as expected for a dopamine agonist. However, these studies did not indicate direct or indirect harmful effects with respect to male fertility.

Adverse Effects/ Undesirable Effects

Expected adverse effects

The following adverse reactions are expected under the use of pramipexole: abnormal dreams, amnesia, behavioural symptoms of impulse control disorders and compulsions such as binge eating, compulsive shopping, hypersexuality and pathological gambling; cardiac failure, confusion, constipation, delusion, dizziness, dyskinesia, dyspnoea, fatigue, hallucinations, headache, hiccups, hyperkinesia, hyperphagia, hypotension, inappropriate antidiuretic hormone secretion, insomnia, libido disorders, nausea, paranoia, peripheral oedema, pneumonia, pruritus, rash and other hypersensitivity; restlessness, somnolence, sudden onset of sleep, syncope, visual impairment including diplopia, vision blurred and visual acuity reduced, vomiting, weight decrease including decreased appetite, weight increase.

Tables 1 and 2 display the adverse reactions reported in patients treated with pramipexole. However, the majority of common adverse reactions were mild to moderate.

Parkinson's disease, most common adverse effects.

The most commonly reported adverse reactions in patients with Parkinson's disease more frequent with pramipexole treatment were nausea, dyskinesia, hypotension, dizziness, somnolence, insomnia, constipation, hallucination, headache and fatigue. More frequent adverse drug reaction in combination with levodopa was dyskinesia. Hypotension may occur at the beginning of treatment, especially if pramipexole is titrated too fast.

Table 1: Parkinson's Disease

System Organ Class	Adverse Drug Reaction
Infections and infestations	
Uncommon	pneumonia
Endocrine disorders	
Uncommon	inappropriate antidiuretic hormone secretion
Psychiatric disorders	
Common	abnormal dreams, behavioural symptoms of impulse control disorders and compulsions, confusion, hallucinations, insomnia
Uncommon	binge eating, compulsive shopping, delusion, hyperphagia, hypersexuality, libido disorder, paranoia, pathological gambling, restlessness, delirium
Rare	Mania
Nervous system disorders	
Very Common	dizziness, impairment of voluntary movement, somnolence
Common	Headache
Uncommon	amnesia, hyperkinesia, sudden onset of sleep, syncope
Eye disorders	
Common	visual impairment including diplopia, vision blurred and visual acuity reduced
Cardiac disorders	
Common	cardiac failure
Vascular disorders	
Common	hypotension
Respiratory, thoracic, and mediastinal disorders	
Uncommon	dyspnoea, hiccups
Gastrointestinal disorders	
Very common	nausea
Common	constipation, vomiting
Skin and subcutaneous tissue disorders	
Uncommon	hypersensitivity, pruritus, rash
General disorders and administration site conditions	
Common	Fatigue, peripheral oedema.

Investigations	
common	weight decrease including decreased appetite
Uncommon	weight increase

Other indication, most common adverse effects

The most commonly reported adverse reactions in patients with Parkinson's disease more frequent with pramipexole treatment were nausea, headache, dizziness and fatigue. Nausea and fatigue were more often reported in female patients compared to males.

Table 2: Other indication

System Organ Class	Adverse Drug Reaction
Infections and infestations	
Uncommon	pneumonia
Endocrine disorders	
Uncommon	inappropriate antidiuretic hormone secretion
Psychiatric disorders	
Common	abnormal dreams, insomnia
Uncommon	behavioural symptoms of impulse control disorders and compulsions such as binge eating, compulsive shopping, hypersexuality, and pathological gambling; confusion, delusion hallucinations, hyperphagia, libido disorder, paranoia, restlessness, mania, delirium
Nervous system disorders	
Common	dizziness, headache, somnolence
Uncommon	amnesia, dyskinesia, hyperkinesia, sudden onset of sleep, syncope
Eye disorders	
UnCommon	visual impairment including diplopia, vision blurred and visual acuity reduced
Cardiac disorders	
UnCommon	cardiac failure
Vascular disorders	
UnCommon	hypotension
Respiratory, thoracic, and mediastinal disorders	
Uncommon	dyspnoea, hiccups
Gastrointestinal disorders	
Very common	nausea
Common	constipation, vomiting
Skin and subcutaneous tissue disorders	
Uncommon	hypersensitivity, pruritus, rash
General disorders and administration site conditions	

Common	Fatigue
Uncommon	peripheral oedema.
Investigations	
Uncommon	Weight decrease including decreased appetite, weight increase

Somnolence

Pramipexole is associated with somnolence and has been associated very rarely with excessive daytime somnolence and sudden sleep onset episodes.

Libido disorders

Pramipexole may be associated with libido disorders (increased or decreased).

Impulse control disorders

Pathological gambling, increased libido, hypersexuality, compulsive spending or buying, binge eating and compulsive eating can occur in patients treated with dopamine agonists including Pramipexole.

Cardiac failure

Pramipexole use was associated with an increased risk of cardiac failure compared with non-use of pramipexole.

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product.

Effect in ability to drive and use machines

Patients being treated with Pramison and presenting with somnolence and/or sudden sleep episodes must be informed to refrain from driving or engaging in activities where impaired alertness may put themselves or others at risk of serious injury or death (e.g. operating machines) until such recurrent episodes and somnolence have resolved.

Pramison is associated with somnolence and has been associated very rarely with excessive daytime somnolence and sudden sleep onset episodes.

Overdose and Treatment

Symptoms:

The expected adverse events should be those related to the pharmacodynamic profile of a dopamine agonist including nausea, vomiting, hyperkinesia, hallucinations, agitation and hypotension.

Treatment:

There is no established antidote for overdosage of a dopamine agonist. If signs of central nervous system stimulation are present, a neuroleptic agent may be indicated. Management of the overdose may require general supportive measures along with gastric lavage, IV fluids and electrocardiogram monitoring.

Haemodialysis has not been shown to be helpful.

Storage Conditions

Keep out of the reach and sight of children.

Do not Store above 30 °C.

Protect from light

Dosage forms and packaging available

Cold form (Alu –Alu) blister pack with printed carton. Blister pack comprises of triple laminated cold formable film consisting of 25µ Polyamide/ 45 µ Aluminium foil /60 µ PVC as the forming material and 25 µ Aluminium foil with 7gsm heat seal lacquer coating as the lidding material. These blisters will be further packed in preprinted carton with package leaflet, according to the approved pack size.

Manufacturer

Aurobindo Pharma Limited (Unit – VII),

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Date of revision

30/05/2019